

Binary Search Tree - Insert Remove

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Q) ID number: 12521464

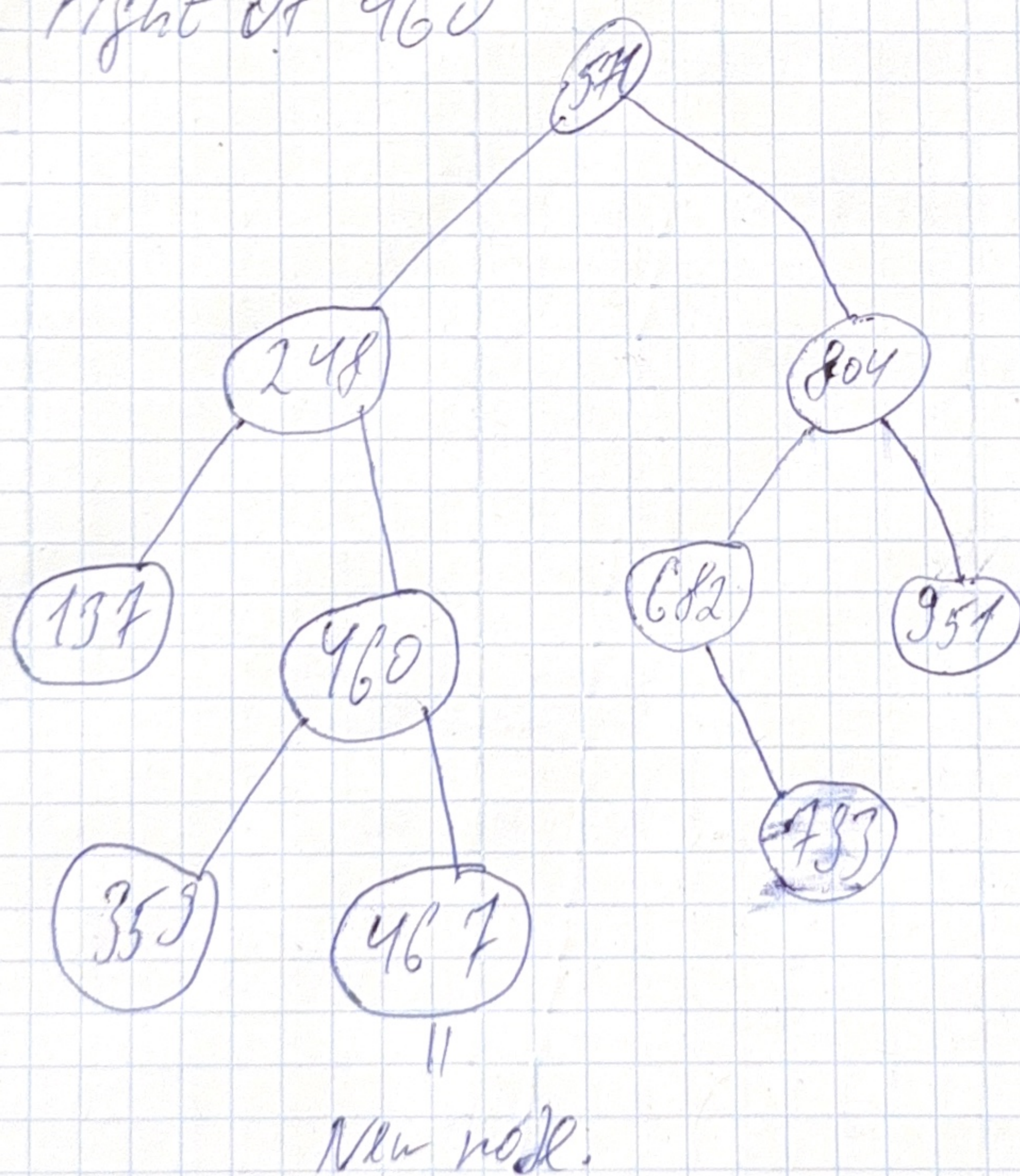
To insert 467 we compare every node with the given one and start from the root going downwards. Therefore:

1. $467 < 541 \Rightarrow$ Go left as in the binary search tree smaller children go to the left and bigger to right.

2. ~~467~~ $467 > 248 \Rightarrow$ Go right (same logic)

3. ~~467~~ $467 > 460 \Rightarrow$ Go right

Finally insert a new node with a value 467 to the right of 460



6) Remove root (571):

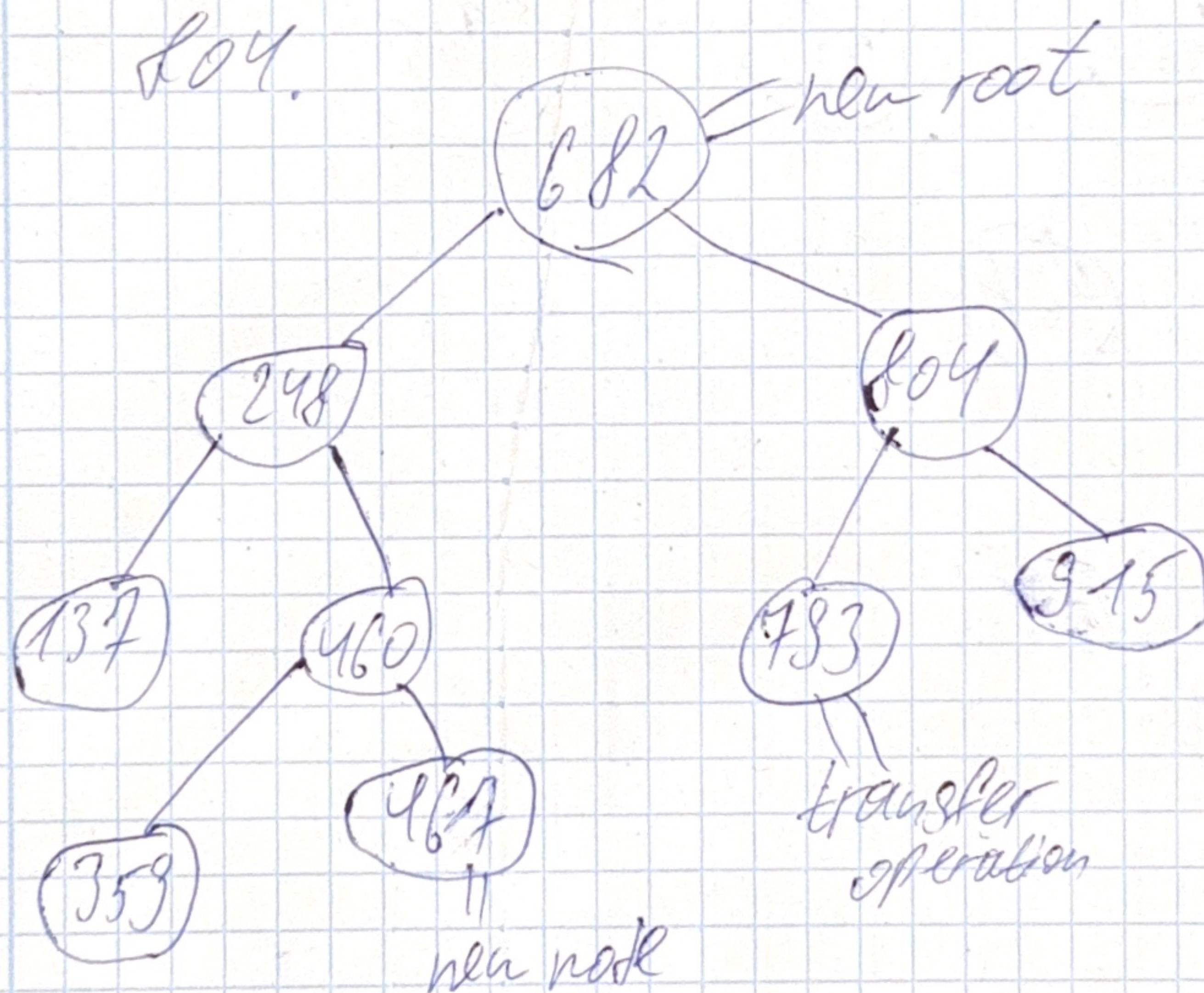
Rules:

We can substitute the root with either: the biggest node from left side; or the smallest from right side.

Therefore: (Apply second option)

1. Go to the right root's child and then to the left most to find the smallest node on the right-hand side. Find 682.

Assign 682 as a new root node and transfer its child 433 instead of its parent as it has no further children, it becomes the child of 804.



C) Insert ID number R 12521467

Number to insert: 125

1. $125 < 682 \Rightarrow$ Go left down

2. $125 < 248 \Rightarrow$ Go left down

3. $125 < 137 \Rightarrow$ Go left down

~~The~~ lower left side of the node 137 is empty, Therefore, inserting the new node 125 as a left child of 137.

